



**YENEPOYA INSTITUTE OF ARTS, SCIENCE, COMMERCE AND
MANAGEMENT**

**A CONSTITUENT UNIT OF YENEPOYA (DEEMED TO BE
UNIVERSITY)**

BALMATTA, MANGALORE

RESEARCH REPORT ON

*Crowd Estimation and Time-Slot Recommendation for Mangalore KSRTC Bus
Services*

SUBMITTED BY

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24MCA306

UNDER THE GUIDANCE OF

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MASTER OF COMPUTER APPLICATION



CERTIFICATE

This is to certify that Leander Chris Dsouza bearing Reg No 24MCA306 has satisfactorily completed the Domain Specific Mini Project Phase 1 Entitled Crowd Estimation and Time-Slot Recommendation for Mangalore KSRTC Bus Services for the 3rd Semester MCA/MSC course during the year 2025-2026.

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1. INTRODUCTION AND BACKGROUND

Public transport plays an important role to meet the daily needs of people across any Indian cities. In Mangalore, KSRTC buses are widely used by students, office commuters, inter-district passengers etc. but there is a need to address the challenges of overcrowding during certain days especially at peak hours as it affects comfort and planning.

Most schedules do not give information about suitable travel timings or crowd levels. Also, existing studies focused only on real-time data, passenger counts, or automated systems which are not always accessible. As a result, there is a need for a simple approach that uses the publicly available data.

This research uses KSRTC Mangalore bus timetables and district-level indicators to analyse the travel patterns and estimate the crowd risk pattern at different time slots. This can help passengers to choose the less congested timings without making real-time claims.

2. LITERATURE SURVEY

- [1]. Basha, SK. et al. (2025): This work involved training and validating models with timestamps, route numbers, and traffic conditions, and comparing performance to improve passenger boarding demand in bus transit by using SVM and DNNs, to forecast demand dynamically. AI-based models delivered high precision in predicting boarding demand.
- [2]. Shrivastava, A. et al. (2024): This work applied transit segment relations (TSR) to capture ridership patterns and predicted crowding levels for different periods of the day. The model, validated with three months of ticketing data in Bhubaneswar, can guide passengers and transit agencies about expected crowding, helping them plan route choices.
- [3]. V S, D. et al. (2022): This work involved IoT, hardware sensors, and deep learning models to estimate crowd density and provide travel information. Such systems can contribute to better use of travel time and help both passengers and travel agencies schedule appropriately.
- [4]. Hsu, WY. et al. (2020): This work processed visual or sensor data to detect head regions and count passengers, evaluating performance across different scenarios. The model performed better than existing methods in estimating occupancy.
- [5]. Marques, S. et al. (2022): This work compared models with and without spatial dependence, finding that models recognizing spatial differences provide better fit and improve predictions. It emphasized the importance of incorporating spatial context and demographic predictors into ridership estimates.
- [6]. Pronello, C. et al. (2024): This work involved installing sniffing devices at stops, collecting probe request data, and training deep learning models to predict passenger counts with an average error of about 1.2 people. Low-cost sensing combined with ML yielded reliable counts for planning.
- [7]. Talusan, JP. et al. (2022): This work used XGBoost and LSTM on millions of observations to perform trip- and stop-level prediction, showing that data fusion and advanced models outperform baseline statistics. Multi-source datasets improve accuracy.
- [8]. Martí, P. et al. (2024): This study introduced a machine learning framework with multi-agent simulations to predict bus ridership and analyze transportation scenarios, using a comprehensive urban bus dataset to forecast passenger flows and evaluate service effectiveness.
- [9]. Fontes, T. et al. (2020): This work applied a deep learning neural network (Multilayer Perceptron) to predict bus passenger demand based on transit ridership and weather conditions over a year, showing that incorporating meteorological variables reduced prediction errors significantly. It labelled ridership data with hour of day and day of week, improving demand forecasting accuracy.
- [10]. Farahmand, Z. H. et al. (2023): This research used deep learning (Multilayer Perceptron networks) to forecast bus ridership accounting for weather, holidays, and special events using smart-card data in the Twente region of the Netherlands, finding that including weather features improves prediction accuracy on regular weekdays.

- [11]. Yusuf, O. et al. (2025): This study developed a horizon-agnostic modelling framework using automated passenger count (APC) data and machine learning to forecast stop-level transit passenger counts, incorporating feature optimization and interpretable model outputs.
- [12]. J Siswanto et al. (2024): This work proposed a deep learning LSTM model for predicting passenger numbers across four bus public transportation operators using a multi-region historical dataset from Auckland Transport, demonstrating how temporal LSTM models can capture passenger flow dynamics for future months.
- [13]. Drisilda, S. et al. (2025): This paper presents a machine learning-based framework to predict hourly ridership at Bangalore Metropolitan Transport Corporation stops using historical bus stop data and SARIMAX models to capture daily patterns and autocorrelations.
- [14]. NNS Amir et al. (2025): This research developed a ridership prediction system for rapid bus services in Kuantan and Penang using multi-feature approaches that integrate route and service variables to forecast passenger demand.
- [15]. Mahsa Movaghar et al. (2025): This study proposed a machine learning framework to estimate ridership loss in the bus network of Stockholm during external crises (e.g., pandemic) by comparing regression and algorithmic models for historical ridership data.
- [16]. Zeb, M. S. et al. (2024): This research applied LSTM-based models to forecast public transit ridership amidst COVID-19 by integrating seasonal, intervention, and pandemic statistical inputs, achieving high predictive accuracy.
- [17]. Akgüngör, A. P., et al. (2025): This work compares multiple deep learning models, including LSTM and NARX, for modelling short-term passenger flows in bus and rail systems using weather and school data, finding that NARX produced the lowest forecast errors.
- [18]. Torrepadula, F. R. di., et al. (2024): This article surveys machine learning methods for public transportation demand forecasting, summarizing varied ML solutions and data types used across studies.
- [19]. Ke, L., et al. (2025): The work emphasizes the importance of having detailed ridership datasets for benchmarking and improving forecasting, scheduling and clustering of bus services. By providing route details, stop distances and transaction-level time granularity, the dataset supports high-resolution analysis of ridership patterns and to evaluate predictive models more precisely.
- [20]. Yamamura, T. et al. (2023): This paper proposes a multiple-LSTM model that integrates past ridership, day of week, time section, weather, and precipitation to improve bus ridership prediction accuracy, demonstrating significant error reduction compared to existing methods.

3. PROBLEM STATEMENT

People using KSRTC buses depend on schedules to plan their journey. Though these give clear information about the route and the timings, they do not give the safest time periods or the crowd levels. As a result, people have to struggle with peak-hour congestion and crowded travel times.

Simultaneously, detailed passenger count data, real-time systems etc. are not publicly available, creating a gap between the required information for travel planning and the data accessible for analysis. There is a need for an approach where publicly available timetable data can be used to estimate the crowd levels and support time-based travel recommendations for KSRTC bus services in Mangalore.

4. SOCIAL RELEVANCE AND UNIQUENESS OF THE WORK

Public transport plays an important role in improving mobility, reducing traffic congestion, and ensuring access to daily services. However, overcrowding in buses often lead to discomfort and safety concerns. This research is socially relevant as it focuses on understanding and predicting bus passenger patterns using data-driven approaches, which can support better planning, improved service quality, and enhanced commuter experience.

The uniqueness of this work lies in its **timetable-driven and population-aware approach** to understand crowd patterns. This research uses **secondary data**, making it low-cost, transparent, and suitable. By linking scheduled bus timings with district-level population and urbanisation indicators, the study provides a practical framework that can be easily adopted by transport authorities, researchers, and media platforms without privacy or operational concerns.

5. AIM, OBJECTIVE AND HYPOTHESIS

Aim

To analyse KSRTC Mangalore bus timetables to estimate route-wise crowd levels and recommend optimal travel time slots for passengers, enabling them to avoid peak-hour congestion and make informed travel decisions.

Objective

- a) To structure KSRTC Mangalore bus timetable data in Microsoft Excel, taking origin, destination, service type, via place, and departure times.
- b) To classify departure times into predefined time slots and integrate district-level population and urbanisation indicators to estimate relative crowd levels.
- c) To develop a framework to assign overcrowding risk levels to different routes and time slots based on structured secondary data.
- d) To create a Tableau-based dashboard to visualise estimated crowd levels and recommend less congested travel time slots for passengers.
- e) To provide insights that are practical, transparent, and reproducible using publicly available data without depending on real-time measurements or sensitive operational information.

Hypothesis

H_0 (Null Hypothesis): There is no significant variation in crowd levels across different KSRTC bus routes and time slots in Mangalore.

H_1 (Alternative Hypothesis): Crowd levels vary across different bus routes and time slots, and structured timetable data combined with population and urbanisation indicators can identify comparatively less crowded travel periods.

6. METHODOLOGY

6.1 Sample Size

The research considers **all KSRTC Mangalore bus routes** listed in the official timetable, including both intra city, inter-state and inter-district services. Since the study uses **secondary data only**, every available route in the timetable is treated as part of the sample.

6.2 Procedure for data collection

- a) KSRTC Mangalore bus timetables will be collected from **official sources**, including bus stand notice boards, KSRTC website, and publicly available PDF schedules.
- b) Timetable data will be **structured in Microsoft Excel**, capturing the following fields: From, To, Service Type, Via Place, Departure Time.
- c) Departure times will be grouped into **predefined time slots** to reflect different periods of daily travel demand in 24-hour format.
- d) District-level **population data (in millions) and urbanisation levels** will be incorporated from official government statistics and mapped to routes based on service coverage.

6.3 Sources and Tools

Primary Data: Not applicable (no field observations or real-time measurements).

Secondary Data:

- a) KSRTC Mangalore official timetables (bus stand notices, PDF schedules, or website)
- b) Government census and urbanisation data for population indicators.

Microsoft Excel: For structuring data, categorisation, and initial calculations.

Python (Pandas, NumPy, optionally scikit-learn): For data cleaning, feature engineering, and classification of crowd risk levels.

Tableau: For creating an interactive dashboard that visualises route-wise crowd levels and recommends suitable travel time slots.

7. REFERENCE

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